WAZP

MONOPHONIC SYNTHESIZER

Owner's Manual

Version 1.0.0



TICCO ROSS | ticcoross.gumroad.com

Welcome

WAZP is a monophonic synthesizer plug-in inspired by the small, buzzy British synths of the late 1970s: two raw oscillators, a screaming state-variable filter, a control oscillator for wobbles and random bleeps, and two envelopes that can repeat on their own. It started life as the WOSP Max for Live device and has been rebuilt as a stand-alone instrument.



Formats

- macOS: Audio Unit, VST3 and standalone app (Apple Silicon and Intel), one .pkg installer.
- Windows: VST3 and standalone app, one Setup.exe installer.
- Linux: VST3, LV2 and standalone app (.tar.gz with an install script).

Installing

macOS: open *WAZP-1.0.0-macOS.pkg* and follow the steps. Click **Customize** to choose formats. The installer is not notarised, so if macOS blocks it: right-click the .pkg and choose **Open**, or allow it in System Settings → Privacy & Security.

Windows: run *WAZP-1.0.0-Windows-Setup.exe*. The VST3 goes to *C:\Program Files\Common Files\VST3*.

Linux: unpack the .tar.gz and run *./install.sh*.

Then rescan plug-ins in your DAW (Ableton Live: Settings → Plug-Ins → Rescan) and load WAZP on a MIDI track.

> WAZP is monophonic: one note at a time, last-note priority. Hold a note and play another for a legato line; with GLIDE up the pitch slides between them.

The Panel

Drag a knob up or down to turn it (scroll wheel works too), double-click to return it to its default. While you move a control, the preset window shows its name and value. The window can be resized from the bottom-right corner.

Keyboard

Control	What it does
SYNC	Hard-syncs oscillator 2 to oscillator 1. Set oscillator 2 higher than oscillator 1 (range or PITCH) and sweep PITCH for the classic tearing sync sound.
GLIDE	Portamento: how long the pitch takes to slide to a new note. Fully left = off, up to about 2 seconds.
Keys	The on-panel keyboard is playable with the mouse (C3 - C5) and lights up with incoming MIDI notes. MIDI pitch bend (+/- 2 semitones) and the mod wheel (adds pitch modulation) also work.

Oscillators

Control	What it does
FT. (both)	Footage / octave range: 32′, 16′, 8′, 4′, 2′. 8′ plays at the pitch of the key.
WIDTH %	Pulse width of oscillator 1, from 10 % to 90 % (50 % is a square wave). Only heard with the pulse waveform.
Wave (both)	OFF, pulse (⊓) or sawtooth. OFF silences the oscillator.
PITCH	Tunes oscillator 2 against oscillator 1, +/- 12 semitones. Small offsets give a fat detuned sound; 7 = a fifth, 12 = an octave.

Mixer / Output

Control	What it does
VOLUME	Master output level, -80 dB to +6 dB.
OSC 1, OSC 2	Level of each oscillator going into the filter (0 - 10). Push both up for more filter drive and grit.

The Panel (continued)

Control Osc.

Control	What it does
FREQ.	Speed of the control oscillator (LFO), 0.05 Hz to 50 Hz.
PITCH MOD.	How much the control oscillator moves the pitch of both oscillators (up to +/- 12 semitones). Vibrato at low settings, sirens and bleeps higher up.
NOISE SIGNAL	Level of white noise in the mixer (0 - 10). Use it alone with the filter for wind, surf and percussion.
Wave	Shape of the control oscillator: sine, square, noise (smoothed random movement) and RND (a new random value every cycle: sample and hold).

Filter

Control	What it does
FREQ.	Filter cutoff frequency, 20 Hz to 20 kHz.
Q	Resonance. High settings make the filter ring and whistle; at the top it almost self-oscillates.
CONTROL	Filter type: LO (low-pass), BAND (band-pass) or HI (high-pass).
Left -/+ knob	Amount of the CONTROL ENV. on the cutoff. Right of centre opens the filter, left of centre closes it (inverted envelope).
Right -/+ knob	Amount of the control oscillator on the cutoff (wah, wobble, random filter steps). Centre = no modulation.

Envelope Generators

Control	What it does
VCA ENV. ATTACK / DECAY	How fast the sound rises when you press a key and how fast it falls to the sustain level. DECAY also sets the release time after you let go.
REPEAT / SUSTAIN LEVEL	Fully left (REPEAT) the envelope keeps re-triggering itself while a key is held: automatic repeating notes. Turned right, it sets the level the sound holds at while the key is down.
CONTROL ENV. ATTACK / DECAY	The envelope that sweeps the filter (through the left -/+ knob). It decays to zero.
REPEAT / DELAY	Fully left (REPEAT) the control envelope loops while the key is held (rhythmic filter sweeps). Turned right, it delays the start of the control envelope by up to 2 seconds.

Presets

The preset window at the top right shows the current preset number and name.

Control	What it does
< >	Step through the factory presets, followed by your own presets.
SAVE	Saves the current sound as a .wazp file. By default it goes to *Documents / Ticco Ross / WAZP / Presets*; presets saved there appear in the < > list.
LOAD	Opens any .wazp file.

WAZP comes with 16 factory presets: INIT, BUZZ BASS, WASP STING, SYNC SCREAM, REPEAT PERC, DREXY ACID, NOISE WIND, SIREN, RANDOM BLEEPS, HOLLOW LEAD, HIGH PASS FIZZ, BAND PLUCK, SUB DRONE, WOBBLE, SLOW SWELL and 8-BIT SQUARE. The presets also appear in your DAW's program list.

> Your DAW saves the complete state (every knob and the preset name) with the project, so you don't need to save a preset to keep a sound in a song.

Automation & MIDI

All controls can be automated and MIDI-mapped from your DAW. WAZP responds to MIDI notes on any channel, pitch bend (+/- 2 semitones) and the mod wheel (CC 1, adds pitch modulation from the control oscillator). Velocity is ignored, as on the original instrument.

Sound Design Tips

- **Classic buzz:** both oscillators on sawtooth, PITCH slightly off zero, Q around 50 %, a short CONTROL ENV. on the filter.
- **Sync lead:** SYNC on, oscillator 2 one range higher, then automate PITCH.
- **Self-playing rhythm:** VCA REPEAT fully left with a short DECAY gives repeating notes; add CONTROL ENV. REPEAT for moving filter pulses.
- **Bleeps:** control-osc wave on RND, FREQ. around 6 - 10 Hz, PITCH MOD. up.
- **Noise percussion:** oscillators OFF, NOISE SIGNAL up, BAND filter with high Q, short VCA decay.

About & Support

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Store: **ticcoross.gumroad.com** Contact: **ticcoross@gmail.com**